	Sanjivani Rural Educational Society's **SANJIVANI COLLEGE OF ENGINEERING** **(An Autonomous Institution)** Kopargaon – 423 603, Maharashtra.	**ACAD-F-15 K**
Academic Year: 2026-27	**One Page Abstract**	**Revision : 00** **Dated : ________**
Department :	**Information Technology Engineering**	**Date of Preparation : 10/07/2026**
Course Code & Name:	**Applied machine Learning for Industry Solutions PECO311C**	**Year/Sem: V**

1. **<u>Activity Title</u>:** Smart Attendance Anomaly Detection Using Machine Learning

2. **<u>Problem Definition</u>:**

   Traditional attendance systems only record whether a student is present or absent. They cannot identify suspicious attendance patterns such as proxy attendance, unusual attendance behavior, or irregular records, which may affect the accuracy and reliability of attendance management.

3. **<u>Abstract</u>:**

   Smart Attendance Anomaly Detection is a machine learning-based system that analyzes attendance records to detect unusual or suspicious patterns automatically. The system studies attendance data such as date, time, subject, and student presence to identify anomalies like sudden attendance changes, proxy attendance, or inconsistent records. This helps educational institutions improve attendance monitoring, reduce manual verification, and maintain accurate attendance records.

4. **<u>Objectives:</u>**
   - To automate attendance anomaly detection.
   - To identify suspicious attendance patterns.
   - To improve the accuracy of attendance records.
   - To reduce manual effort in attendance verification.

5. **<u>Input:</u>**
   - Student attendance records
   - Student ID
   - Date and time
   - Subject details
   - Attendance status (Present/Absent)

6. **Operations / Overview:**
• Collect attendance data.
• Preprocess the dataset by removing missing values.
• Apply a Machine Learning anomaly detection algorithm (such as Isolation Forest or Local Outlier Factor).
• Detect abnormal attendance behaviour.
• Generate alerts and reports for detected anomalies.

7. **Expected Output:**
• Collect attendance data.
• Preprocess the dataset by removing missing values.
• Apply a Machine Learning anomaly detection algorithm.
• Detect abnormal attendance behaviour.
• Generate alerts and reports for detected anomalies.

8. **Applications:**
• Schools and Colleges
• Universities
• Coaching Institutes
• Corporate Employee Attendance Systems

9. **Scope:**
The system can be integrated with biometric devices, RFID cards, facial recognition, or mobile-based attendance systems. Future improvements can include AI-based prediction of attendance trends and real-time anomaly alerts.

10. **References:**

• Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*, O'Reilly.
• Scikit-learn Documentation – Anomaly Detection.
• IEEE Research Papers on Attendance Management Systems.
• Kaggle Attendance Datasets.

Submitted by:

Roll No	PRN No	Name of the Student	Signature
05	**0124UITM1005**	**Swarup Pingle**	
51	**0124UITM1051**	**Abhijeet Raghuvanshi**	
52	**0124UITM1052**	**Shubham Khedkar**	
60	**0124UITM1060**	**Kshitij Shinde**	

Course In charge